ROBUSTFOLIO · 7 ways to spread money across 18 US ETFs, 168 out-of-sample months

PRE-REGISTERED, MECHANICALLY EVALUATED, NON-CONFIRMATORY | NET OF 10 bps ONE-WAY COSTS

I tried to break my own result before I published it.

Four defences against fooling yourself, one that failed, and a null I can stand behind.

THE FOUR DEFENCES, AND THE ONE THAT FAILED

Freeze before looking.

Metric, thresholds and decision rule dated before the run — so the benchmark could not be chosen after the answer was known.

Prove no leakage mechanically.

A test suite rather than a code review, and one of its tests is recorded as vacuous rather than passed.

Certify the numerics.

1,176 solves, zero rejections, worst KKT residual 1.49e-11 against a 1e-7 acceptance bar.

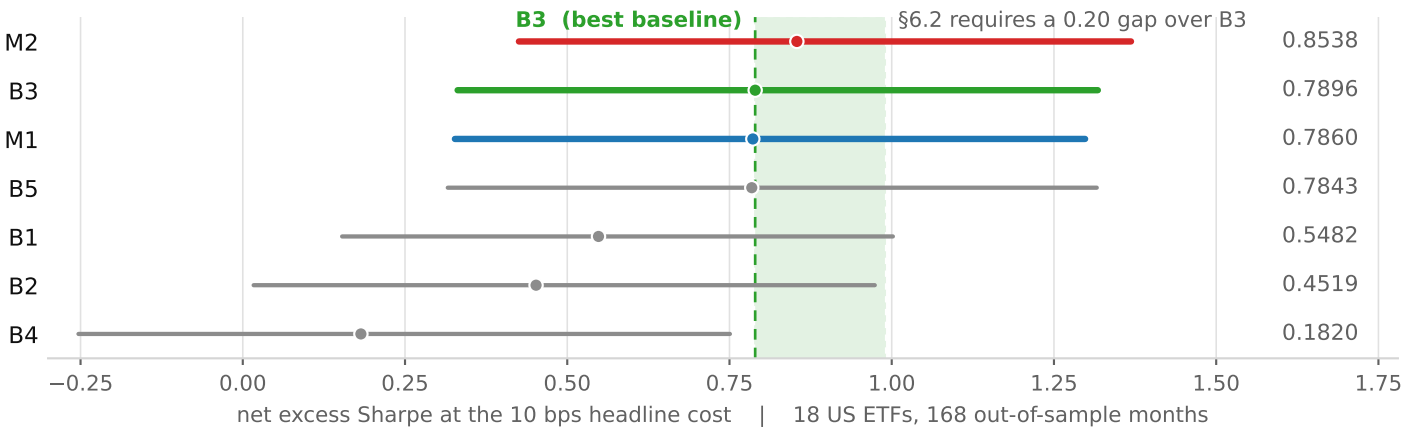
Require significance AND materiality AND robustness.

Both candidates pass a p-value against zero (0.0032 and 0.0017) and fail the one against the best baseline (0.495 and 0.587 against a bar of 0.95). The rule requires both.

THE DEFENCE THAT FAILED

Validating the report generator required running the frozen configuration — which WAS the confirmatory configuration. The registered test is re-designated NON-CONFIRMATORY: the hypothesis is untested, not falsified.

THE NULL UNDERNEATH IT



EXPLORATORY AND NON-CONFIRMATORY. Net excess Sharpe at the 10 bps headline cost: dot at the point estimate, bar at its 90% bootstrap interval. No confirmatory test exists, so the hypotheses are untested rather than falsified. Not an investable return.

AND THE REPRODUCTION CLAIM THAT FAILED TWICE

A claim of bit-for-bit reproduction — mine — failed twice. 4.2e-17 on one column: a BLAS kernel picks its summation order at run time from the CPU's features. 4.698e-12 on thirteen: a solver's output is a property of (data, code, machine), not of (data, code).

The claim is now split: decisions compared exactly, measurements bounded, and the observed difference printed on every run.

Don't trust the summary — run the check. verify_release.py prints EXACT or WITHIN, and the verdict it compared. github.com/kira-ml/robustfolio

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